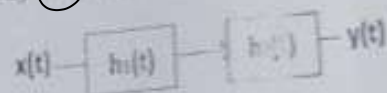


Student Name:

Student ID:

Q1. Select the correct answer for the following questions:

1. The result of multiplying $x(t) = t^2$ and $g(t) = \sin(3t)$ is..... function.
a. Odd b. Even c. Neither even nor odd. d. Can not be determined.
2. The convolution between $x(t) = e^{-t}u(t-4)$ and $\delta(t+4)$ for $t > 1$ is given as.....
a. $e^{-4t}u(-4)$ b. e^{-4} c. $e^{-4t}u(t-8)$ d. $\frac{1}{e^{4(t+1)}}$ e. $e^{4t}u(t-8)$ f. $e^{4t-8}u(t)$
3. For LTI system, a system is a system.
a. Memoryless, causal. b. Time-invertible, causal. c. Stable, memoryless. d. Causal, stable. e. Linear, stable.
4. The overall impulse response $h(t)$ of the following cascade LTI system, if $h_1(t) = \delta(t-X)$ and $h_2(t) = \delta(t-5)$, is given as...
a. $h(t) = \delta(t-2)$ b. $h(t) = \delta(t+2)$ c. $h(t) = \delta(t-2)$ d. $h(t) = \delta(t-8)$ e. $h(t) = \delta(t-5)$



5. One of the following signals is considered as causal system, which is
a. $y[n] = 2x[n] - 3x[n+1]$
b. $y[n] = 2x[n-7] - 3x[n+1]$
c. $y[n] = 2x[n] + 3x[n+1]$
d. $y[n] = 2x[n] + 3$
e. None of them

6. The result of $\cos(2t) + \cos(-2t)$ is.....
a. $2\cos(2t)$ b. $0.5\cos(t)$ c. $2\cos(t)$ d. $2\cos(4t)$ e. $0.5\cos(2t)$ f. None of them

7. The following signal $x[n] = \sin(\frac{2\pi n}{5})$ is considered as

- a. Periodic, discrete, and $N = 5/2$
- b. Periodic, discrete, and $N = 5$
- c. Aperiodic, continuous, and $N = 5/2$
- d. Aperiodic, discrete, and $N = 2/5$
- e. Aperiodic, discrete and $N = 5$

8. The system is considered as a memory system.

- a. $y(n) = x(n) + x(n+1)$
- b. $y(t) = x^2(t) + x(t)$
- c. $y(t) = x(t)x(t-1)$
- d. a and b
- e. a and c

9. Which of the following is NOT true?

- a. $t^2 \delta(t) = 0$
- b. $\cos(t) \delta(t) = \delta(t)$
- c. $\sin(t) \delta(t - \pi) = \delta(t - \pi)$
- d. $\int_{-\infty}^{\infty} (2t^2 + 1) \delta(t) dt = 1$

10. The result of $\int_0^4 3\delta(t+1) dt$ is given as.....

- a. 3
- b. $3t$
- c. $12t$
- d. 0
- e. 12

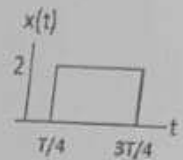
11. The power of the signal $s(t) = 6\sin(5\pi t - \frac{\pi}{4}) + 4\cos(17\pi t)$ is.....

- a. 10
- b. 100
- c. 52
- d. 26
- e. None of them

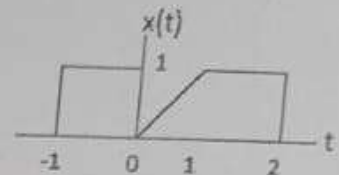
12. The fundamental period of the signal $x(t) = 2\sin\left(\frac{8t}{2}\right) + \cos(4t/6 + 40^\circ)$ is.....
 a. 4π b. 24π c. $\pi/4$ d. $6\pi/8$ e. $4\pi/2$ f. None of them

13. The signal $x(t)$ that represents the following figure is.....

a. $-2u\left(t - \frac{T}{2}\right) + 4u\left(t - T/2\right)$ b. $2\text{rect}\left(\frac{t - \frac{T}{4}}{T}\right)$ c. $2\text{rect}\left(\frac{t - \frac{T}{2}}{T}\right)$ d. $2\text{rect}\left(\frac{t}{T} - \frac{1}{4}\right)$



Q2. Draw the even component of the signal $x(t)$ shown below. (Show all necessary steps)



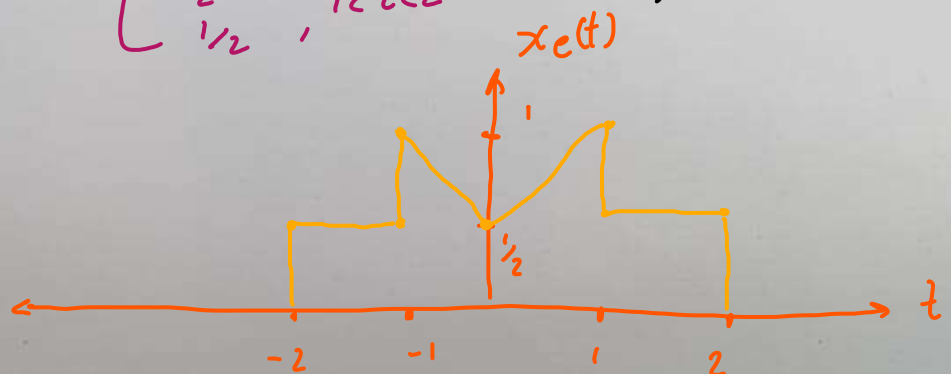
$$x(t) = \begin{cases} 1, & -1 < t < 0 \\ t, & 0 < t < 1 \\ 1, & 1 < t < 2 \\ 0, & \text{o.w.} \end{cases}$$

$$x(-t) = \begin{cases} 1, & -1 < -t < 0 \\ -t, & 0 < -t < 1 \\ 1, & 1 < -t < 2 \\ 0, & \text{o.w.} \end{cases}$$

$$\Rightarrow x(-t) = \begin{cases} 1, & -2 < t < -1 \\ -t, & -1 < t < 0 \\ 1, & 0 < t < 1 \\ 0, & \text{o.w.} \end{cases}$$

$$x_e(t) = \frac{x(t) + x(-t)}{2} = \begin{cases} \frac{1}{2}, & -2 < t < -1 \\ \frac{1-t}{2}, & -1 < t < 0 \\ \frac{1+t}{2}, & 0 < t < 1 \\ \frac{1}{2}, & 1 < t < 2 \end{cases}$$

$(-2, \frac{1}{2}), (-1, \frac{1}{2})$
 $(-1, 1), (0, \frac{1}{2})$
 $(0, \frac{1}{2}), (1, 1)$
 $(1, \frac{1}{2}), (2, \frac{1}{2})$



Q1 Answers:

1	2	3	4	5	6	7	8	9	10	11	12	13

Good Luck!

1) $x(t) = t^2 \rightarrow \text{even}$
 $y(t) = \sin 3t \rightarrow \text{odd}^x$ } $\rightarrow \text{odd}$ A.

2) $x(t) * \delta(t+4) = x(t+4) = y(t)$

$y(t) = e^{-(t+4)} u(t+4-4) = e^{-(t+4)} \text{ for } t > 0.$

$y(t) = e^{-(t+4)} u(t-1) \rightarrow y(t) = \frac{1}{e^{t+4}} \text{ for } t > 1$ D.

3) LTI system : A.

A. Memoryless $\xrightarrow{\text{then}}$ Causal
 (instantaneous) (indep of future). ✓
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B. Time Invertible $\xrightarrow{\text{then}}$ Causal ✗
 $y(t) = x(-t)$ $y(-1) = x(1)$

C. Stable $\xrightarrow{\text{then}}$ memoryless

$h(t) = e^t u(t)$

$|y(t)| \leq M |h(t)|$

BIBO.

$y(t) = x(t) * h(t)$ ✗
 $= \int_{-\infty}^{\infty} x(\tau) e^{t-\tau} u(t-\tau) d\tau$
 has memory $\leftarrow t \geq \tau$

⋮

$$4) h(t) = h_1(t) * h_2(t)$$

$$= \int_{-\infty}^{\infty} \delta(\tau-3) \delta(t-5-\tau) d\tau \quad E.$$

$$= \delta(t-5-3) = \delta(t-8).$$

5) causal system :-

$$D. y[0] = 2x[0] + 3$$

$$6) \cos 2t + \cos(-2t) = \cos 2t + \cos 2t = 2\cos 2t. \quad A.$$

$$7) x[n] = \sin\left(\frac{2\pi}{5}t\right) \quad B.$$

discrete, periodic, $N = \frac{2\pi}{\omega_0} = \frac{2\pi}{1} \times \frac{5}{2\pi} = 5.$

8) memory system :- $E. A \& C.$

$$A. y(0) = x(0) + x(1), \quad C. y(0) = x(0) x(-1).$$

9) NOT True: $\sin t \delta(t-\pi) = \cancel{\sin \pi} \delta(t-\pi)$
 $= \delta(t-\pi) \neq \delta(t-\pi) \quad C.$

$$10) \int_0^4 3\delta(t-(-1)) dt = 0 \rightarrow t=-1 \notin [0,4], \quad D.$$

$$11) \quad P \{ A \sin(\omega t + \theta) \} = A^2/2$$

$$P \{ A \cos(\omega t + \theta) \} = A^2/2$$

$$P_{tot} = \frac{6^2}{2} + \frac{4^2}{2} = 18 + 8 = 26. \quad D.$$

$$12) \quad x(t) = 2 \sin(4t) + \cos\left(\frac{2}{3}t + 40^\circ\right)$$

$$T_1 = \frac{2\pi}{4} = \frac{\pi}{2}, \quad T_2 = \frac{2\pi}{1} \times \frac{3}{2} = 3\pi$$

$$\frac{T_1}{T_2} = \frac{\pi}{2} \times \frac{1}{3\pi} = \frac{1}{6} \longrightarrow T_2 = T_0$$

$T_0 = 3\pi$, therefore f. none of the above.

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13) All answers are wrong.

$$x(t) = 2 \operatorname{rect}\left(\frac{t - T/2}{T/2}\right).$$

closest answer is C.